SUPPORTING ADULT GUIDANCE

Measuring and drawing angles

Angles Tell Us Why · Lesson 1. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Turning Arm, learners turn an arm from a fixed base arm (by dragging, with buttons, or with the arrow keys). The turn is shaded and named live, and an on-screen protractor can be laid on it, with the scale that starts at 0 on the base arm in red. Learners change the arm length and turn the whole drawing to see that neither changes the angle. Eight mystery angles follow: the number is hidden, learners estimate the type first, then measure; reading the other scale is caught and explained. Six questions and a stretch on reflex angles follow. Learners then make body turns, judge statements, and make angles by eye.

Driving Question: *How do we measure a turn?*

The mathematics, in plain terms

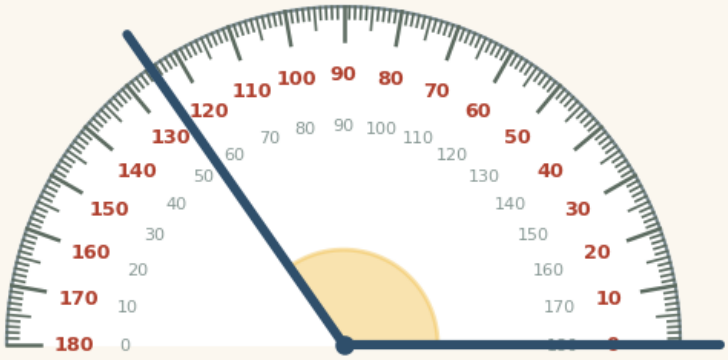
An angle is a measure of rotation; a full turn is 360°. The two-scale protractor is the main source of error: learners read the wrong scale (55 for 125). Estimating first, using 90° and 180° as landmarks, prevents most of these errors. Reflex angles are measured indirectly as 360° minus the angle on the other side. Angles on a straight line and at a point come in Lesson 2.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Measuring an obtuse angle

How big is this angle?

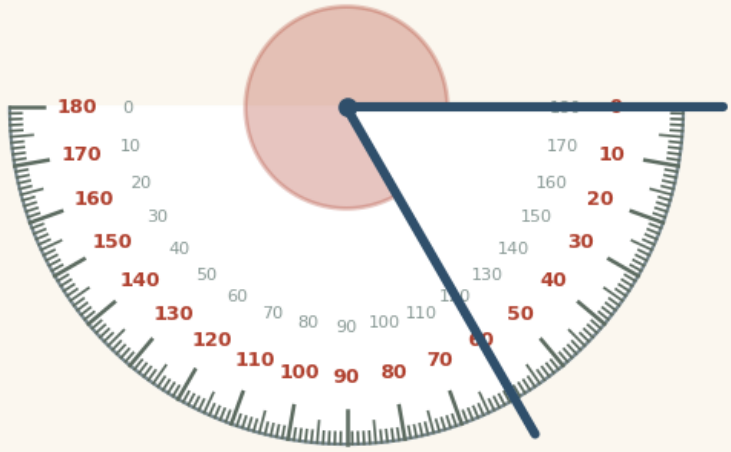


1. Estimate: more than a right angle, less than a straight line.	obtuse
2. The red scale starts at 0 on the base arm. Where does the other arm cross it?	125
3. The grey scale shows 55 there, but 55° would be acute.	not 55
4. Check against the estimate.	125° is obtuse ✓

The angle is 125°, an obtuse angle.

Example 2 — Measuring a reflex angle

How big is the shaded reflex angle?



1. Estimate: past a straight line, so	reflex
2. The protractor reaches only 180°. Flip it below the line and measure the turn the other way.	60°
3. The two turns together make a full turn.	360°
4. So the reflex angle is	360° – 60° = 300°

The reflex angle is 300°.

Turn first, then measure, then judge by eye

Learners see an angle as a turn before using a protractor, so the protractor measures something they already understand. Estimating always comes before measuring. Body turns link degrees to fractions of a full turn; judging by eye builds the sense of size that makes estimates reliable.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Turning Arm, mystery angles and six questions. Stretch: measuring reflex angles.	~16 min
Movement	Turn yourself: quarter, half, three-quarter and full turns.	~7 min
Reflection	True or not? Six statements about angles.	~7 min

Creativity	Make it by eye: angles without numbers.	~10 min
Rest	Turns are everywhere.	~2 min
Nutrition	Intellectual: checking.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Reading the wrong scale (55° for an obtuse angle of 125°).
Ask: “Is the angle more or less than a right angle? Which reading fits?”

Thinking longer arms make a bigger angle.
Ask: “If we make the arms shorter, has the turn changed?”

Not placing the protractor's centre exactly on the vertex.
Ask: “Where is the point of the turn? Where is the protractor's centre?”

Thinking an angle changes when the page is turned.
Ask: “Turn the page. Has the gap between the arms changed?”

Trying to read a reflex angle straight from a 180° protractor.
Ask: “What is the turn the other way? What do the two make together?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: right angles and quarter turns. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner measures an obtuse angle and reads 55°. The aim is a question that lets them catch it with an estimate.

- “Before reading, is this angle bigger or smaller than a right angle?”
- “Which scale starts at 0 on the arm you lined up?”

- “Does 55° match what you can see?”
- “What does the other scale show?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Do learners need a real protractor?
It helps for the Final Task, but the lesson works without one. If a real one is used, check that the centre sits on the vertex.

Should learners measure to the nearest degree?
Yes for the main activity (the lesson accepts a reading within 1°). On paper, within 2° is reasonable.

Is the stretch required?
No. The secure core is estimating, measuring and classifying angles up to 180°.

What comes next?
Lesson 2: angles on a straight line and at a point.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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